Social Media Engagement Analytics

Objective

To analyze social media engagement data using Python and derive meaningful insights related to:

- Post performance (likes, comments, shares)
- Engagement rate trends across content types
- Audience behavior across countries and devices
- Sentiment-based engagement analysis
- Feature engineering (engagement score, hashtag count)

Database Description

This dataset contains social media post interaction data with 5000 records.
It captures:

- User and post details (user_id, post_id, post_type, post_category)
- User demographics (age, gender, country)
- Engagement metrics (likes, comments, shares, engagement_rate)
- Content context (hashtags, sentiment, device_type)
- Behavioral metrics (watch_time_sec, impression_count, follower_count)

Tables and Structure

Column Name	Data Type	Description
user_id	int64	Unique ID of user
age	float64	Age of user
gender	object	Gender of user
country	object	User country
post_id	int64	Unique post ID
post_type	object	Type of post
post_category	object	Content category
likes	float64	Number of likes
comments	float64	Number of comments
shares	float64	Number of shares
watch_time_sec	int64	Watch time in seconds
impression_count	int64	Number of impressions
posted_at	object	Timestamp of post
follower_count	int64	Number of followers
is_verified	bool	Verified account status
device_type	object	Device used
sentiment	object	Sentiment label
hashtags	object	Hashtags used
engagement_rate	float64	Engagement percentage

Engineered Features

Feature	Description
Engagement_Score	likes + comments + shares
hashtag_count	Number of hashtags
Likes_Category	Categorized likes (Low/Medium/High)
Comments_Category	Categorized comments
Shares_Category	Categorized shares
Ranking	Rank based on engagement score

Tools and Technologies

Pandas

- Data loading, cleaning, transformation, and analysis
- Functions used: read_csv(), groupby(), cut(), fillna(), value_counts()

NumPy

- Numerical operations and mathematical functions
- Used for log transformation and numerical calculations

Matplotlib

- Basic data visualization
- Used for: bar charts, line plots, histograms, box plots, scatter plots

Seaborn

- Advanced statistical visualization
- Used for: heatmaps, count plots, violin plots, swarm plots, pair plots

Plotly

- Interactive visualizations
- Used for dynamic charts like scatter plots and bar plots

Key Insights

Content Performance

Post Type vs Engagement

- Different post types show **different engagement levels**
- Usually, **video-based content (Reels/Videos)** performs better than static posts
- Image/text posts tend to have **moderate or lower engagement**

Best Performing Content Category

- Some content categories consistently get higher engagement rates
- These categories indicate what users prefer most (e.g., entertainment, education, lifestyle)

Country-wise Engagement

- Engagement rate is not uniform across countries
- Some countries show very high interaction levels, while others are lower

Social Media Engagement Analysis

Countries with Highest Average Engagement Rate

- Engagement is not evenly distributed across countries
- A few countries consistently show higher interaction levels
- These regions likely have: More active users , Better content relevance
- Higher mobile usage

User Trends — Age vs Engagement

- Younger users generally show higher engagement rates
- Engagement tends to decrease slightly with increasing age
- Middle-age users show moderate but stable interaction

Verified vs Non-Verified Accounts

- Verified accounts usually get:
- Higher impressions
- Better engagement rate
- But in some cases, non-verified accounts also perform well if content is strong

Behavioral Insights

Best Time of Day for Impressions

- Engagement is higher during:
- Morning peak hours
- Evening active hours
- Lower activity during late night hours

Device Type vs Watch Time

- Mobile users show higher watch time
- Desktop users have lower but more stable viewing behavior

Sentiment Analysis

Positive sentiment posts perform best

- They receive:
- Higher likes
- More shares
- Better engagement rate

Negative vs Neutral Sentiment Behavior

- Negative posts: May get attention but lower engagement quality
- Neutral posts: Consistent but moderate performance
- Both underperform compared to positive sentiment

Conclusion

- Focus on video/reel content to increase engagement
- Prioritize high-performing post categories for content strategy
- Target countries with highest engagement rates for better reach
- Post during peak activity hours to maximize impressions
- Optimize content for mobile users (highest watch time)
- Encourage positive sentiment content for better performance
- Strengthen strategies for young audience (high engagement group)
- Maintain consistency but focus more on quality over quantity
- Use verified accounts to improve credibility and reach
- Track and improve engagement score regularly for performance monitoring

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